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AGRICULTURAL AND BIOSYSTEMS ENGINEER LICENSURE EXAMINATION

AREA III: *AB Structures, Environment Engineering, Bioprocess Engineering and Allied Subjects*

PREPARED BY: **ENGR. MELANIE D. PIEDAD**
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1. In psychrometric chart, these represented by lines at diagonals.
 - a. Dry Bulb Temperature
 - b. Wet Bulb Temperature
 - c. Specific Enthalpy
 - d. Specific Volume
2. This law states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.
 - a. 1st Law of Thermodynamics
 - b. 2nd Law of Thermodynamics
 - c. 3rd Law of Thermodynamics
 - d. 0th Law of Thermodynamics
3. In PAES 421:2009, What is the minimum carriage way of one-lane and two-lane FMR, excluding shoulders?
 - a. 2-4 m
 - b. 4-6 m
 - c. 2.5-5 m
 - d. 3-6 m
4. In PAES 402:2000, the recommended brooding temperature for 8-14 days chicks.
 - a. provide heat when necessary
 - b. 29-32 °C
 - c. 32-35 °C
 - d. 27-29 °C
5. What is the defect of wood that causes abnormal growth which occurs at the starting point of a limb or branch of a tree?
 - a. shake
 - b. knot
 - c. check
 - d. wane
6. In PAES 410:2000, the lairage shall be provided with lighting intensity of _____ lux.
 - a. 200
 - b. 100
 - c. 500
 - d. 110
7. What diode is needed as protection against lightning in solar installations?
 - a. Bypass diode
 - b. Blocking diode
 - c. Shading diode
 - d. Lightning diode
8. In PAES 403:2001, under the individual nest subsection, there shall be one nest for each ____ to ____ layer.
 - a. 5 - 6
 - c. 7 - 8

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b. 8 - 10

d. 4 - 6

9. In CPES rating, the maximum weight rating for Environmental, Safety and Health criteria during construction.

a. 0.40

c. 0.07

b. 0.15

d. 0.30

10. A complete lighting unit consisting of a lamp or lamps together with parts designed to distribute the light, to position and protect the lamps, and to connect the lamps to the power supply.

a. lumen

c. luminous

b. luminaire

d. lighting fixture

11. The primary law governing blackbody radiation that quantifies the intensity of radiation emitted by unit surface area into a fixed direction from the blackbody as a function of wavelength for a fixed temperature.

a. Stefan-Boltzmann Law

c. Fourier's Law

b. Newton's Law of Cooling

d. Planck's Law

12. In CPES rating, what is the percentage for time upon completion?

a. 0.10

c. 0.40

b. 0.15

d. 0.30

13. What is the size of the wire AWG 10?

a. 5.0 mm²

c. 6.0 mm²

b. 5.5 mm²

d. 8.8 mm²

14. Amount of heat added or removed measured in change of temperature without changing state.

a. Latent Heat

c. Specific Heat

b. Sensible Heat

d. Innate Heat

15. Measures the ability of the machine to remove hulls with minimum breakage.

a. Coefficient of wholeness

c. Hulling Efficiency

b. Hulling Coefficient

d. Fineness Modulus

16. The amount paid to project owners from Clean Development Mechanism (CDM) fund for carbon emission reduction projects.

a. Carbon credit value

c. Carbon trading value

b. Emission value

d. Carbon emission value

17. What is the peak amperage of a solar battery charging station equipped with 16 units of 75 watt-peak 4.16-ampere solar module? The system is divided into 4 charging channels, each channel has 4 modules connected in parallel.

a. 32.2 A

c. 16.6 A

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b. 25.4 A

d. 8.3 A

18. What is the minimum methane content of biogas in order to operate biogas-fueled generators?

a. 50%

c. 85%

b. 65%

d. 90%

19. Type of biogas plant which have aerobic conditions in the upper layers and anaerobic processes occurring in the bottom layers, especially in the settled solids?

a. Aerobic Biogas

c. Facultative Biogas

b. Anaerobic Biogas

d. Activated Biogas

20. The wastewater, either treated or untreated, discharged from a reservoir, basin or treatment plant.

a. Effluent

c. Turbid Water

b. Influent

d. Hard Water

21. In PAES 421:2009, the tank shall have heating device able to heat a complete charge of bituminous liquid to ____ °C.

a. 80

c. 180

b. 120

d. 200

22. In PAES 415:2001, under natural ventilating systems of greenhouse, large vent openings provide the most ventilation with total vent area that should be ____ - ____ of floor area.

a. 15% - 30%

c. 25% - 50%

b. 30% - 50%

d. 15% - 25%

23. In PAES 421:2009, it is constructed to drain water from the subgrade and to collect surface water either from the roadway surface or adjacent.

a. ditches

c. channel

b. culvert

d. conduit

24. In PAES 408:2001, the minimum space requirement for carabao feedlot feeding shall be ____ m/animal.

a. 0.5

c. 0.75

b. 1.0

d. 1.25

25. The maximum number of outlets for a 15-A circuit.

a. 8-10 outlets

c. 14-15 outlets

b. 10-12 outlets

d. 6-8 outlets

26. Dissolved oxygen concentrations will be the lowest within a waterbody on ____.

a. Afternoon

c. Sunset

b. Sunrise

d. Noon

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27. In PAES 401:2000, heaters shall be installed in creep areas to provide newborn pigs its required temperature of ____°C - ____°C until they are 3 days old.

- a. 32-35
- b. 29-35
- c. 30-35
- d. 27-35

28. What is the mechanical power of a windmill with a diameter of 10 meters and velocity of 5 mps?

- a. 55.92 watts
- b. 981.75 watts
- c. 3,926.99 watts
- d. 9,327.56 watts

29. A Carnot engine is working between two reservoirs with temperatures of 1000°C and 500°C, what is the thermal efficiency of the system?

- a. 10%
- b. 30%
- c. 50%
- d. 100%

30. Environmental-friendly refrigerant

- a. CFC
- b. HFC
- c. HCFC
- d. FC

31. It is the programming language of Arduino microcontrollers.

- a. R
- b. Python
- c. C/C++
- d. Java

32. In Arduino programming, the meaning of IDE is:

- a. Integrated Development Environment
- b. Internal Development Environment
- c. International Development Environment
- d. Interconnected Development Environment

33. The Feed-in-Tariff (FIT) of solar energy in the BiGSHOW is:

- a. 6.63 PHP/kW-hr
- b. 9.68 PHP/kW-hr
- c. 5.90 PHP/kW-hr
- d. 8.53 PHP/kW-hr

34. It is the gas produced from agricultural waste which is odorless, tasteless, colorless and flammable

- a. CH₄
- b. CO
- c. O₂
- d. CO₂

35. All are products of pyrolysis except:

- a. Vinegar
- b. Tar
- c. Oil
- d. Syngas

36. What is the electrical energy consumption of a 15-kW vermicast production shredder plugged in a 220-V source if it requires 2 hrs to shred 1.5 tonnes of farm waste?

- a. 0.20 kW-h/kg
- b. 0.02 kW-h/kg
- c. 0.002 kW-h/kg
- d. 0.022 kW-h/kg

37. It is a type of whitening machine consisting of a ribbed cylinder

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enclosed in a perforated steel housing

- a. friction type
- b. abrasive type
- c. rubber roll type
- d. cone "cono" type

38. What is the input current to a 5 hp single phase electric motor use to drive a hammer mill at 2200 rpm. The input voltage is 220 volt. Assume a power factor of 0.98

- a. 14.5 amp
- b. 17.3 amp
- c. 20.4 amp
- d. None of the above

39. When a system deviates infinitesimally from equilibrium at every instant of its state, it is undergoing:

- a. isobaric process
- b. quasi-steady process
- c. isometric process
- d. isentropic process

40. A rice thresher that is fed with panicles at one end and discharges threshed product at the other end is called ____.

- a. radial type thresher
- b. hold-on type thresher
- c. axial-flow thresher
- d. cross-flow thresher

41. A wooden pole (SG=0.65), 9 cm by 9 cm by 5 m long, hangs vertically from a string in such a way that 3 m is submerged in water and 2 m is above the surface. What is the tension in the string?

- a. 19.87 N
- b. 17.89 N
- c. 18.96 N
- d. 16.97 N

42. How much energy is needed for a 5-volt source to move 4 coulombs of charge?

- a. 20 joules
- b. 1.25 joules
- c. 0.8 joules
- d. 80 joules

43. It is a hypothetical activity which takes no resource or time to complete. It is represented by broken arrowed line & is used for either distinguishing activities having common starting & finishing events or to identify & maintain proper precedence relationship between activities that are not connected by events

- a. Free Float
- b. Slack Activity
- c. Dummy activity
- d. Zero Slack

44. In lateral and sub-lateral canal where the available roadway width is 3 meters, the carriageway width shall be 2 meters and the shoulder width shall be 0.5 meters. If the available roadway width is 4 meters, the shoulder shall expand to 1 meter.

- a. road bed lateral canal
- b. roadway on main irrigation
- c. roadway on lateral & sub lateral irrigation canals
- d. Turn-out section canal

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45. When you want to use objects from a drawing file in another application, you can cut these objects to the Clipboard and then paste them into another application. CUTCLIP copies selected objects to the Clipboard, removing them from the drawing using the shortcut:

- a. Ctrl Y
- b. Ctrl Z
- c. Ctrl X
- d. Ctrl V

46. In AutoCAD, it creates copies of objects arranged in a pattern. You can create copies of objects in a regularly spaced rectangular, polar, or path

- a. Ortho
- b. Offset
- c. Array
- d. Fillet

47. A structure the consists of compressive force action and supports only axial loadings is:

- a. Beam
- b. Column
- c. Purlins
- d. Girder

48. A beam needs 4 pcs of 20 mm RSB. If there is no available 20 mm RSB, how many 16 mm RSB can substitute for 20 mm?

- a. 4 pcs
- b. 5 pcs
- c. 6 pcs
- d. 7 pcs

49. How many pieces of umbrella nails are there if there are 10 kilograms of umbrella nails?

- a. 10,000 pcs
- b. 11,000 pcs
- c. 12,000 pcs
- d. 12,500 pcs

50. It a positively charged particle made up of two protons and two neutrons and released by a radioactive nucleus

- a. Alpha radiation
- b. Beta radiation
- c. Ultra-violet radiation
- d. Gamma radiation

51. Given the initial temperature of 45 F and initial pressure of 16 bar, the pressure changed to 35 bar, determine the change in temperature.

- a. 250
- b. 300
- c. 500
- d. 600

52. It is the production of food, fiber, or other plant or animal products using farming techniques that protect the environment, public health, human communities, and animal welfare.

- a. Organic Agriculture
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

53. It is a production system that sustains the health of soils, ecosystems and people. It relies on ecological processes,

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biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects.

- a. Organic Agriculture
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

54. It is the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the Subbase or if no Subbase, upon the sub-grade. Treatment may include application of chemical-based soil additives such as soil-stabilizers and/or any approved method.

- a. Aggregate Base Course
- b. Base Course
- c. Subbase Course
- d. Aggregate Subbase Course

55. It involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions

- a. Aerated static pile
- b. In-vessel system composting
- c. Windrow composting
- d. Bin composting

56. The percent of moisture content of test materials for peanut sheller.

- a. 8%-10%, wet basis
- b. 12%-14%, wet basis
- c. 8%-10%, dry basis
- d. 12%-14%, dry basis

57. Measures velocity by the change of source light or sound frequency from the frequency of reflections from moving particles.

- a. Optical Strobe
- b. Doppler
- c. Bellow
- d. Platometer

58. Ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent.

- a. Scattering loss
- b. Cylinder loss
- c. Separation loss
- d. Blower loss

59. A 2.4-tonne capacity *Jatropha* oil expeller was loaded full with dried *Jatropha* seeds in 4 minutes. It produced 600 kg of crude oil in 2 hours. What is its crude oil production rate?

- a. 300 kg/h
- b. 200 kg/min
- c. 600 kg/h
- d. 600 kg/min

60. A 10MW-25MW hydro power plant is classified as: _____.

- a. Micro hydro
- b. Small hydro
- c. Mini hydro
- d. Medium hydro

61. Convert the binary number 010101_2 to decimal number system

- a. 53_{10}
- c. 42_{10}

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b. 85_{10}

d. 21_{10}

62. Which one is an integrated circuit used in instrumentation and control?

a. TD 8062

c. ES8813

b. LM333

d. CQ 556

63. In PAES 421:2009, the minimum radius of curvature for flat, rolling and mountainous terrains respectively shall be.

a. 75, 50 & 5 m

c. 150, 100 & 50 m

b. 500, 150 & 50 m

d. 600, 200 & 100 m

64. It refers to the extent or degree by which the bran layer of the brown rice is removed as a result of whitening

a. Whitening degree

c. Milling recovery

b. Milling degree

d. Milling index

65. The most efficient and ideal gas cycle.

a. Rankine Cycle

c. Otto Cycle

b. Carnot Cycle

d. Brayton Cycle

66. Standard thickness of concrete of trapezoidal ditch w/ riprap

a. 0.25 m

c. 0.30 m

b. 0.15 m

d. 0.20 m

67. Lighting computation in a warehouse with 4x4 grid lighting layout with mounting height ratio of 1.0

a. 4 m

c. 2 m

b. 16 m

d. 8 m

68. Contains less than or equal to 20% longitudinal streaks

a. Well-milled rice

c. Overmilled rice

b. Regular milled rice

d. Undermilled rice

69. Moisture gradient of corn and rice in mechanical dryer

a. 0.5%

c. 2.5%

b. 1.0%

d. 2.0%

70. It is the chamber maintained under pressure for uniform distribution of the heated air through the grain mass

a. dust collection system

c. plenum

b. drying chamber

d. tempering bin

71. The rate of radiation of a black body at 0°C is E watt. Then the rate of radiation of this black body at 273°C will be:

a. $8E$

c. $4E$

b. $16E$

d. E

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72. In PAES 420:2002, the door height shall be provided with a clearance of at least _____ m above the tallest machinery

- a. 0.3 m
- b. 0.4 m
- c. 0.6 m
- d. 0.7 m

73. Heat absorbed by a unit mass of a material at its boiling point to convert the material into a gas without temperature change

- a. Latent heat of sublimation
- b. Latent heat of deposition
- c. Latent heat of fusion
- d. Latent heat of vaporization

74. Which of the following is a good insulator

- a. Brass
- b. Mica
- c. Tin
- d. Platinum

75. The angle measured from the horizontal at which product will start to move downwards over a smooth surface with gravity discharging the product is called a.

- a. angle of discharge
- b. angle of friction
- c. angle of repose
- d. hopper angle

76. In PAES 616:2016, precautionary measures in *Wastewater Re-Use for Irrigation*, which is false?

- a. Direct contact with wastewater shall be avoided
- b. Use of fine mist for sprinkler irrigation shall be used
- c. Potable and wastewater lines shall not cross connect
- d. Irrigation with wastewater shall be stopped immediately when algal bloom occurs

77. Corn Grits with 0.86 mm to 1.09 mm length.

- a. Grit #16
- b. Grit #18
- c. Grit #10
- d. Grit #14

78. The amount of moisture in the grain expressed as a percentage of the total weight of the samples.

- a. Moisture content dry basis
- b. Moisture content wet basis
- c. Moisture content
- d. Equilibrium moisture content

79. Recommended stacking density for corn in terms of number of bags per cubic meter.

- a. 10
- b. 9
- c. 12
- d. 15

80. A 1m x 100m poultry handling area is to be lighted at an average maintained footcandle of 50 fc. How many 2-lamp fixtures of 120cm

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long F40 T12 WW rapid start fluorescent lamps are required assuming the CU is 0.35 and the LLF is 0.75?

- a. 3 fixtures
- b. 6 fixtures
- c. 2 fixtures
- d. 4 fixtures

81. The heat exchange when a material changes its state of molecular aggregation without change in temperature.

- a. Latent heat
- b. Specific heat
- c. Sensible heat
- d. All of the above

82. A systematic approach to identifying, evaluating, and controlling food safety hazards.

- a. CPES
- b. HACCP
- c. GMP
- d. SSOP

83. For the purely inductive circuit, the power factor is:

- a. 0
- b. <1
- c. >1
- d. 1

84. What force is required to punch a 20 mm diameter hole in a plate that is 25 mm thick? The shear strength is 350 MPa

- a. 550 kN
- b. 400 kN
- c. 500 kN
- d. 450 kN

85. In PAES 418:2002, area in sqm per ton of commodity in primary processing plant is:

- a. 10 sqm per ton
- b. 15 sqm per ton
- c. 20 sqm per ton
- d. 25 sqm per ton

86. When the surface of an air-cooling evaporator falls below 0°C, frost will form. Frost is:

- a. Air and ice
- b. Air and water
- c. Frozen water
- d. Ice foam

87. A truck with axle loads of 40 kN and 60 kN on a wheel base of 5 m rolls across a 10-m span. Compute the maximum bending moment.

- a. 160 kN·m
- b. 122.5 kN·m
- c. 245 kN·m
- d. 320 kN·m

88. Biochemical Oxygen Demand (BOD₅) in mg/L for wastewater re-use:

- a. <500
- b. <200
- c. <150
- d. <50

89. In an inductive circuit, the inductance is 1.5H. What is the inductive resistance at 50Hz source?

- a. 471 ohms
- b. 416 ohms
- c. 461 ohms
- d. 417 ohms

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90. Corn initially at 24% is to be dried to 14%. If the initial weight of the shelled corn is 1000 kilos, what is its final weight?
- a. 893.72 kg
 - b. 783.72 kg
 - c. 883.72 kg
 - d. 383.72 kg
91. In PAES 411:2000, the gut room/tripery shall be provided with lighting intensity of _____ lux.
- a. 200
 - b. 220
 - c. 540
 - d. 110
92. Final weight for 9 tons of rice to be dried from 24% to 14%.
- a. 7.9 ton
 - b. 8.9 ton
 - c. 9.9 tons
 - d. 10.9 tons
93. In PAES 409:2001, how many milking stalls are required for a milking herd having 21-30 animals with a milking machine?
- a. 1
 - b. 2
 - c. 3
 - d. 4
94. Determine the volume of the digester required to produce biogas for a livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5.
- a. 445 m³
 - b. 469 m³
 - c. 424 m³
 - d. 438 m³
95. What is the equivalent size in mm² of a 250 MCM cable?
- a. 126.67 mm²
 - b. 136.67 mm²
 - c. 146.67 mm²
 - d. 156.67 mm²
96. Burning efficiency of the biomass furnace shall be at least:
- a. 98%
 - b. 95%
 - c. 90%
 - d. 80%
97. Stoichiometric air requirement for pyrolysis
- a. 10-20%
 - b. 20-30%
 - c. 30-40%
 - d. 40-50%
98. The C/N Ratio of agricultural waste
- a. 10:20
 - b. 20:30
 - c. 30:40
 - d. 40:50
99. Main product of gasification
- a. CH₄
 - b. CO
 - c. O₂
 - d. CO₂
100. The heating system efficiency for rice hull direct-fired mechanical grain dryer
- a. 95%
 - b. 80%
 - c. 90%
 - d. 65%

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